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```
clc;
clear;
close all;
```

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DESIGN AND COMPARATIVE ANALYSIS OF BUTTERWORTH, CHEBYSHEV-I AND ELLIPTIC IIR FILTERS ==========

1. INPUT PARAMETERS

```
Fs  = 1000;     % Sampling frequency (Hz)
Fp  = 100;      % Passband edge (Hz)
Fst = 150;      % Stopband edge (Hz)
Rp  = 1;        % Passband ripple (dB)
Rs  = 40;       % Stopband attenuation (dB)

% Normalized frequencies
Wp  = Fp/(Fs/2);
Wst = Fst/(Fs/2);
```

2. CHEBYSHEV TYPE-I FORMULA

```
epsilon = sqrt(10^(Rp/10)-1);

N_formula = acosh(sqrt((10^(Rs/10)-1)/(10^(Rp/10)-1))) ...
            / acosh(Fst/Fp);

N_theory = ceil(N_formula);
```

3. BUTTERWORTH FILTER

```
[Nb,Wb] = buttord(Wp,Wst,Rp,Rs);
[bB,aB] = butter(Nb,Wb,'low');
```

4. CHEBYSHEV TYPE-I FILTER

```
[Nc,Wc] = cheb1ord(Wp,Wst,Rp,Rs);
[bC,aC] = cheby1(Nc,Rp,Wc,'low');
```

5. ELLIPTIC FILTER

```
[Ne,We] = ellipord(Wp,Wst,Rp,Rs);
[bE,aE] = ellip(Ne,Rp,Rs,We,'low');
```

6. DISPLAY INPUT PARAMETERS

```
fprintf('\n=================================================\n');
fprintf('       IIR FILTER COMPARATIVE ANALYSIS\n');
fprintf('=================================================\n');
```

```
fprintf('\nINPUT PARAMETERS\n');
fprintf('Sampling Frequency       = %.2f Hz\n',Fs);
fprintf('Passband Edge Frequency  = %.2f Hz\n',Fp);
fprintf('Stopband Edge Frequency  = %.2f Hz\n',Fst);
fprintf('Passband Ripple          = %.2f dB\n',Rp);
fprintf('Stopband Attenuation     = %.2f dB\n',Rs);
```

```
==================================================
       IIR FILTER COMPARATIVE ANALYSIS
==================================================

INPUT PARAMETERS
Sampling Frequency       = 1000.00 Hz
Passband Edge Frequency  = 100.00 Hz
Stopband Edge Frequency  = 150.00 Hz
Passband Ripple          = 1.00 dB
Stopband Attenuation     = 40.00 dB
```

7. CHEBYSHEV FORMULA RESULTS

```
fprintf('\nCHEBYSHEV TYPE-I FORMULA\n');
fprintf('Ripple factor epsilon    = %.4f\n',epsilon);
fprintf('Calculated order         = %.4f\n',N_formula);
fprintf('Theoretical minimum N    = %d\n',N_theory);
```

```
CHEBYSHEV TYPE-I FORMULA
Ripple factor epsilon    = 0.5088
Calculated order         = 6.2071
Theoretical minimum N    = 7
```

8. FILTER ORDER COMPARISON

```
fprintf('\n=====================================================\n');
fprintf('              FILTER ORDER COMPARISON\n');
fprintf('=====================================================\n');

fprintf('Butterworth Order        = %d\n',Nb);
fprintf('Chebyshev Type-I Order   = %d\n',Nc);
fprintf('Elliptic Order           = %d\n',Ne);
```

```
==================================================
              FILTER ORDER COMPARISON
==================================================
Butterworth Order        = 12
Chebyshev Type-I Order   = 6
Elliptic Order           = 4
```

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9. COEFFICIENTS ==========

```
fprintf('\n==================================================\n');
fprintf('           BUTTERWORTH COEFFICIENTS\n');
fprintf('==================================================\n');

fprintf('Numerator b =\n');
disp(bB);

fprintf('Denominator a =\n');
disp(aB);

fprintf('\n==================================================\n');
fprintf('           CHEBYSHEV TYPE-I COEFFICIENTS\n');
fprintf('==================================================\n');

fprintf('Numerator b =\n');
disp(bC);

fprintf('Denominator a =\n');
disp(aC);

fprintf('\n==================================================\n');
fprintf('             ELLIPTIC COEFFICIENTS\n');
fprintf('==================================================\n');

fprintf('Numerator b =\n');
disp(bE);

fprintf('Denominator a =\n');
disp(aE);
```

```
==================================================
           BUTTERWORTH COEFFICIENTS
==================================================
Numerator b =
   1.0e-03 *

  Columns 1 through 7

    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050
```

```
  Columns 8 through 13

    0.1757    0.1098    0.0488    0.0146    0.0027    0.0002

Denominator a =
  Columns 1 through 7

    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643

  Columns 8 through 13

  -27.6092   11.5218   -3.4662    0.7128   -0.0899    0.0053


==================================================
           CHEBYSHEV TYPE-I COEFFICIENTS
==================================================
Numerator b =
   1.0e-03 *

    0.0464    0.2782    0.6956    0.9274    0.6956    0.2782    0.0464

Denominator a =
    1.0000   -4.8694   10.3807  -12.3367    8.5969   -3.3268    0.5586


==================================================
             ELLIPTIC COEFFICIENTS
==================================================
Numerator b =
    0.0197   -0.0171    0.0333   -0.0171    0.0197

Denominator a =
    1.0000   -3.0330    3.8118   -2.2911    0.5554
```

10. FREQUENCY RESPONSES

```
N = 4096;

[HB,f] = freqz(bB,aB,N,Fs);
[HC,~] = freqz(bC,aC,N,Fs);
[HE,~] = freqz(bE,aE,N,Fs);

MB = 20*log10(abs(HB)+eps);
MC = 20*log10(abs(HC)+eps);
ME = 20*log10(abs(HE)+eps);
```

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11. PASSBAND RIPPLE ==========

```
P = f <= Fp;

Ripple_B = max(MB(P))-min(MB(P));
Ripple_C = max(MC(P))-min(MC(P));
Ripple_E = max(ME(P))-min(ME(P));
```

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12. STOPBAND ATTENUATION ==========

```
S = f >= Fst;

Stop_B = -max(MB(S));
Stop_C = -max(MC(S));
Stop_E = -max(ME(S));
```

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13. PERCENTAGE DEVIATION ==========

```
Dev_C_Ripple = abs(Ripple_C-Rp)/Rp*100;

Dev_B_Stop = abs(Stop_B-Rs)/Rs*100;
Dev_C_Stop = abs(Stop_C-Rs)/Rs*100;
Dev_E_Stop = abs(Stop_E-Rs)/Rs*100;
```

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14. TRANSITION CHARACTERISTICS
==========

```
TransitionBW = Fst-Fp;
```

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15. DISPLAY PERFORMANCE ==========

```
fprintf('\n=====================================================\n');
fprintf('             FILTER PERFORMANCE\n');
fprintf('=====================================================\n');

fprintf('\nPASSBAND RIPPLE\n');
fprintf('Butterworth      = %.4f dB\n',Ripple_B);
fprintf('Chebyshev Type-I = %.4f dB\n',Ripple_C);
fprintf('Elliptic         = %.4f dB\n',Ripple_E);

fprintf('\nSTOPBAND ATTENUATION\n');
fprintf('Butterworth      = %.4f dB\n',Stop_B);
fprintf('Chebyshev Type-I = %.4f dB\n',Stop_C);
fprintf('Elliptic         = %.4f dB\n',Stop_E);

fprintf('\nCHEBYSHEV RIPPLE DEVIATION\n');
fprintf('Specified Ripple = %.2f dB\n',Rp);
fprintf('Measured Ripple  = %.4f dB\n',Ripple_C);
fprintf('Percentage Deviation = %.4f %%\n',Dev_C_Ripple);

fprintf('\nTRANSITION CHARACTERISTICS\n');
fprintf('Passband Edge     = %.2f Hz\n',Fp);
fprintf('Stopband Edge     = %.2f Hz\n',Fst);
fprintf('Transition Band   = %.2f Hz\n',TransitionBW);
```

```
=====================================================
             FILTER PERFORMANCE
=====================================================

PASSBAND RIPPLE
Butterworth      = 0.8033 dB
Chebyshev Type-I = 1.0000 dB
Elliptic         = 1.0000 dB

STOPBAND ATTENUATION
Butterworth      = 40.0198 dB
Chebyshev Type-I = 41.3365 dB
Elliptic         = 40.0000 dB

CHEBYSHEV RIPPLE DEVIATION
Specified Ripple = 1.00 dB
Measured Ripple  = 1.0000 dB
Percentage Deviation = 0.0000 %

TRANSITION CHARACTERISTICS
Passband Edge     = 100.00 Hz
Stopband Edge     = 150.00 Hz
Transition Band   = 50.00 Hz
```

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16. MAGNITUDE RESPONSE ==========

```
figure;

plot(f,MB,'LineWidth',1.5);
hold on;
plot(f,MC,'LineWidth',1.5);
plot(f,ME,'LineWidth',1.5);

grid on;

xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

title('Comparison of IIR Filter Magnitude Responses');

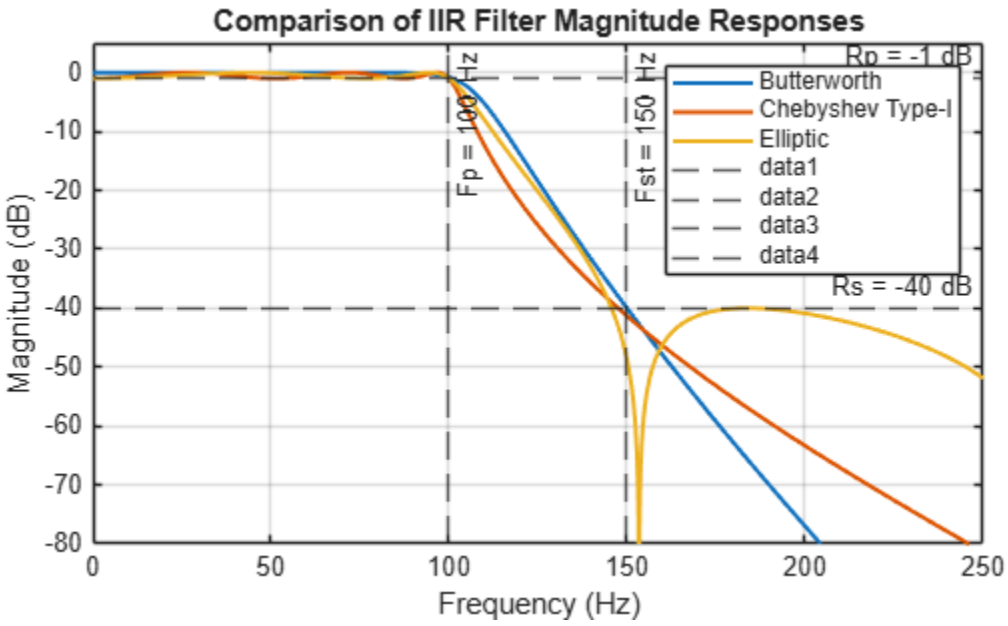
legend('Butterworth','Chebyshev Type-I','Elliptic');

xlim([0 250]);
ylim([-80 5]);

xline(Fp,'--','Fp = 100 Hz');
xline(Fst,'--','Fst = 150 Hz');

yline(-Rp,'--','Rp = -1 dB');
yline(-Rs,'--','Rs = -40 dB');

hold off;
```



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17. PHASE RESPONSE ==========

```
Phase_B = unwrap(angle(HB))*180/pi;
Phase_C = unwrap(angle(HC))*180/pi;
Phase_E = unwrap(angle(HE))*180/pi;

figure;

plot(f,Phase_B,'LineWidth',1.2);
hold on;
plot(f,Phase_C,'LineWidth',1.2);
plot(f,Phase_E,'LineWidth',1.2);

grid on;

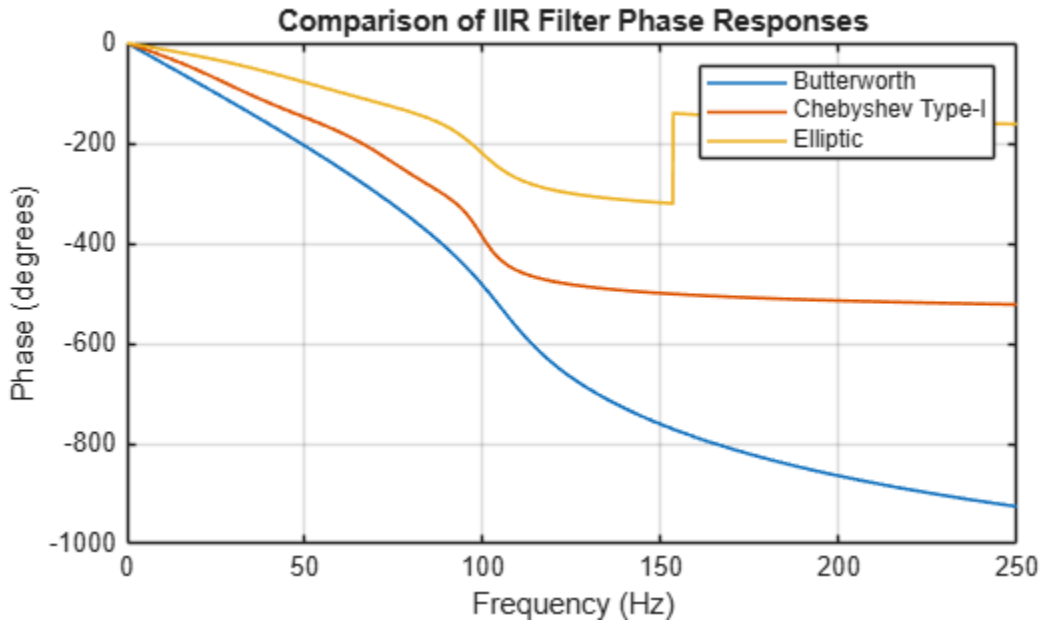
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');

title('Comparison of IIR Filter Phase Responses');

legend('Butterworth','Chebyshev Type-I','Elliptic');

xlim([0 250]);

hold off;
```



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18. FILTER ORDER GRAPH ==========

```
figure;

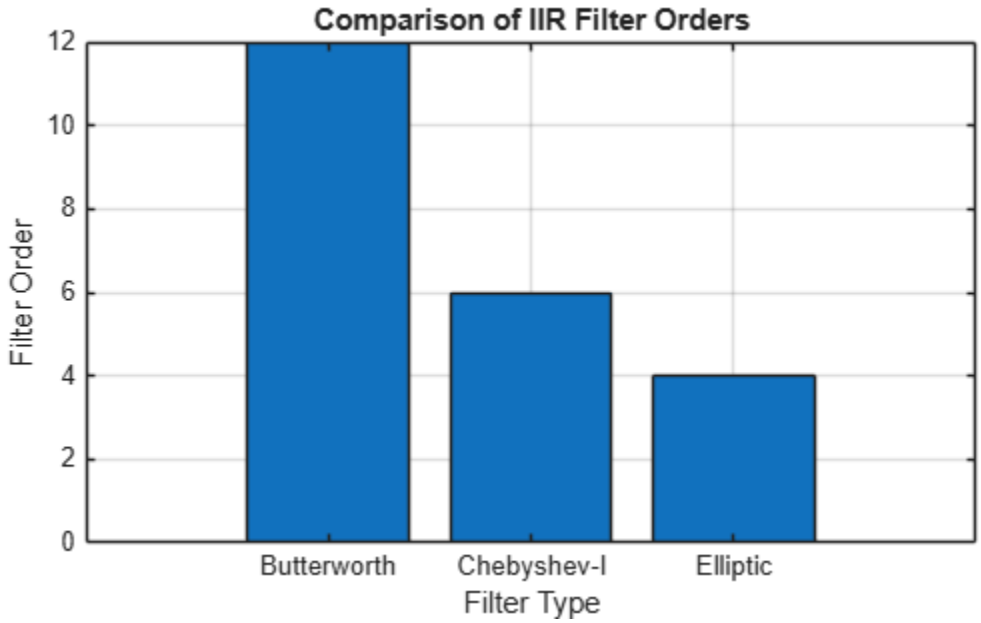
bar([Nb Nc Ne]);

set(gca,'XTick',1:3);
set(gca,'XTickLabel',...
    {'Butterworth','Chebyshev-I','Elliptic'});

ylabel('Filter Order');
xlabel('Filter Type');

title('Comparison of IIR Filter Orders');

grid on;
```



Published with MATLAB® R2026b